Q1) 16 general purpose registers

1 reg is 16 bits

16 instructions → 4 bit opcodes

Ranges :-

16 registers : R0 to R15

Immediate for LDI : -8 to +7 (in 2's complement) 4 bits.

Offset :- -8 to +7 (2's complement, 4 bits)

Branch displacement :- -16 to +15 (2's complement, 5 bits)

Choosing instruction :- 4 bits for unique opcodes.

Rules :

→ LD & ST : (value in Rb, treated as an unsigned number) + (signed offset)

- Address must be b/w 0 to 255 & even

→ BEQ & BNE : (address of next instruction after branch) + (signed displacement in bytes)

→ Target for branch, JMP must be first byte of a valid instruction

LDI : immediate sign extended to 16 bits

Task 1

Opcode ⇒ 4 bits
register filed ⇒ 4 bits

LDI imm ⇒ 4 bits
offset ⇒ 4 bits
disp ⇒ 8 bits

For ADD, SUB :- Instruction length = 16

AND, OR, XOR

4 + 4 + 4 , + 4
Rd, Rs1, Rs2, Opcode

INC, DEC :- 4 + 4 length = 8
R Opcode

MOV :- 4 + 4 + 4 length = 12
Rd Rs Opcode

LDI :- 16 bits (sign extended
If MSB 0's all bits to the left 0
If MSB 1's all bits to the left 1)

LD, ST :- [4], [4] + [4], [4]
Rd, Rs Rb signed offset opcode

length = 16 bits

BNQ, BNE :- [4], [4], [5] [4]
Rs1 Rs2 disp (signed) Opcode

17?

can we use a 3-digit opcode or put it as 0000, 0001, etc.

JMP :- [8 bits], [4]
JMP target opcode

length = 12

RET :- [4], [4]
Rs opcode

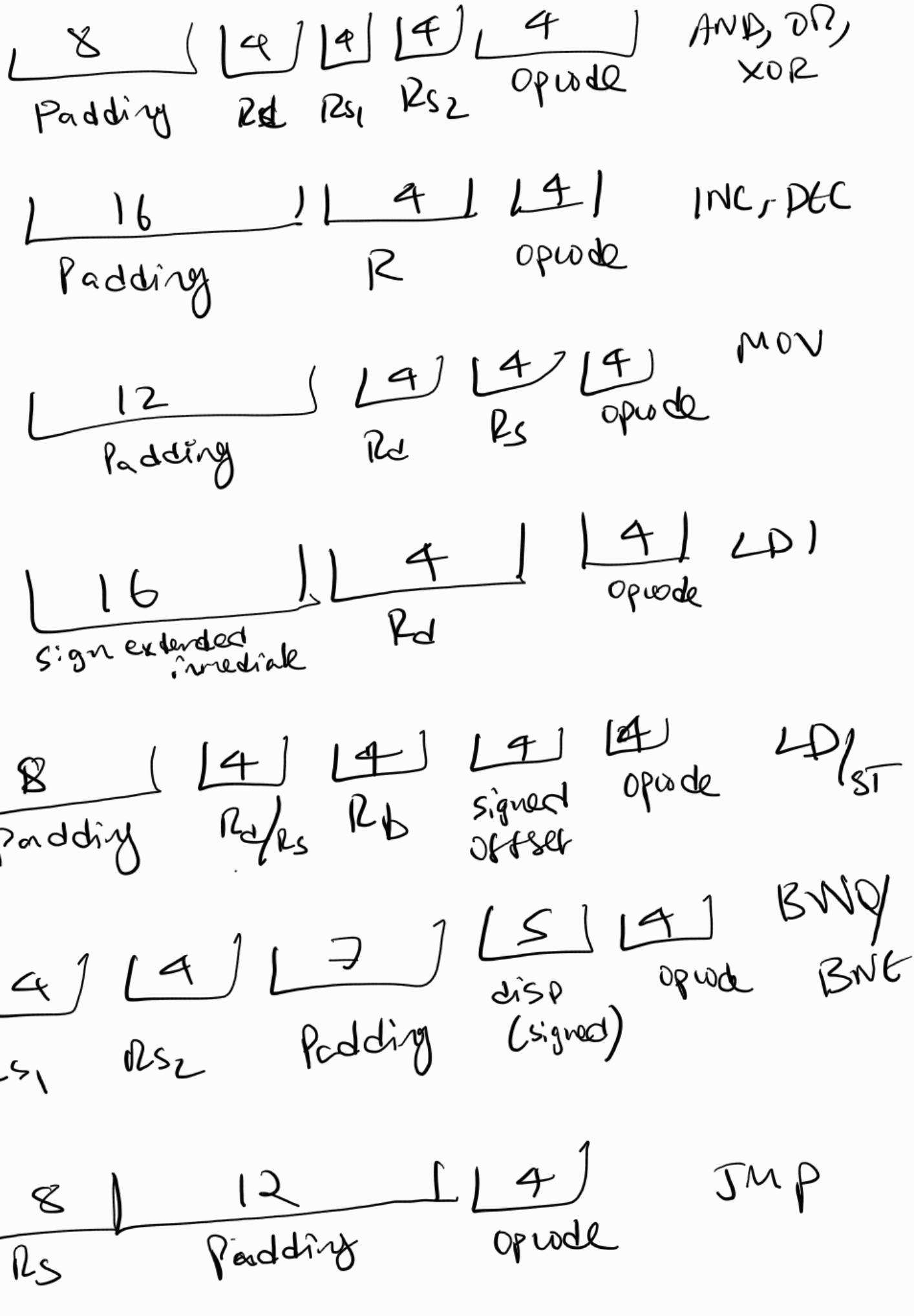
length = 8

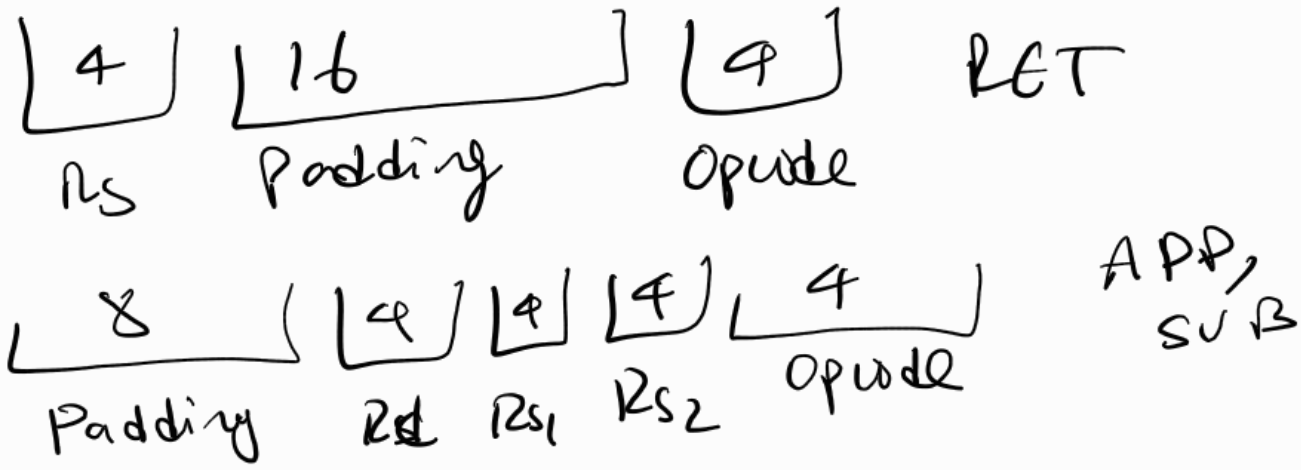
NOP :- [4]
opcode

length = 4

2) We must choose 24 bits as BNQ and BNE require 17 bit instructions We can use padding bits to fill the extra in

3)


8
4
4
4
4
Padding
Rd
Rs1
Rs2
Opcode
AND, OR, XOR
16
4
4
Padding
R
Opcode
INC, DEC
12
4
4
4
Padding
Rd
Rs
Opcode
MOV
16
4
4
LDI
sign extended immediate
Rd
Opcode
8
4
4
4
4
Padding
Rd/Rs
Rb
signed offset
Opcode
LD/ST
4
4
7
5
4
Rs1
Rs2
Padding
disp (signed)
Opcode
BNQ/BNE
8
12
4
Rs
Padding
Opcode
JMP



4) Opcodes

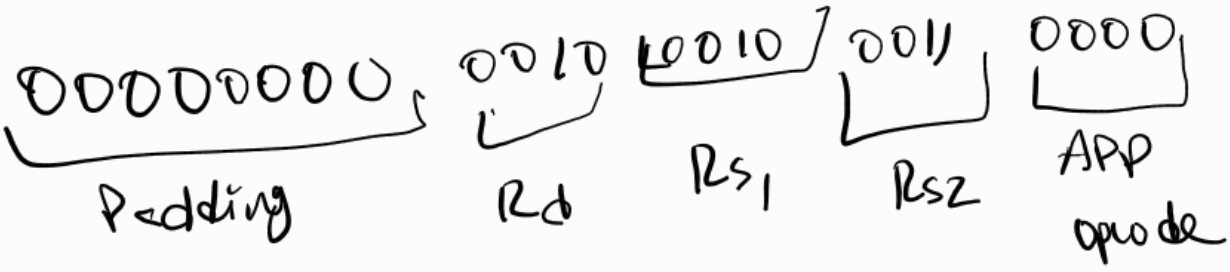
0000	ADD
0001	SUB
0010	AND
0011	OR
0100	XOR
0101	INC
0110	DEC
0111	MOV
1000	CPI
1001	LD
1010	ST
1011	BEQ
1100	BNG
1101	JMP
1110	RET
1111	NOP

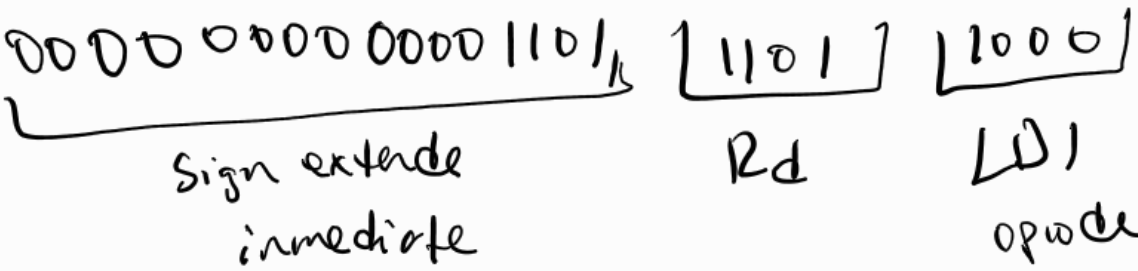
Last 4 digits of every instruction is the opcode so the system knows where to check for what to do every time.

5) a) Every range is accounted for and the padding bits give more flexibility in case something is overlooked

b) Every opcode is unique so no two instructions will ever be the same as data memory addresses are also unique

c) Decoder can look at the last 4 digits of each instruction for the opcode and then figure out where to look for the rest of the fields.

6) a) 

b) 

c) 0000 0000 | 0010 | 1010 | 0001 | 1001

Padding | Rd | Rb | offset | LD opcode

d) 00101011 | 0000 0000 0000 | 1101

JMP target | Padding | JMP opcode